

Real-time diabetic retinopathy screening by deep learning in a multisite national screening programme: a prospective interventional cohort study



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Summary

Background Diabetic retinopathy is a leading cause of preventable blindness, especially in low-income and middle-income countries (LMICs). Deep-learning systems have the potential to enhance diabetic retinopathy screenings in these settings, yet prospective studies assessing their usability and performance are scarce.

Methods We did a prospective interventional cohort study to evaluate the real-world performance and feasibility of deploying a deep-learning system into the health-care system of Thailand. Patients with diabetes and listed on the national diabetes registry, aged 18 years or older, able to have their fundus photograph taken for at least one eye, and due for screening as per the Thai Ministry of Public Health guidelines were eligible for inclusion. Eligible patients were screened with the deep-learning system at nine primary care sites under Thailand's national diabetic retinopathy screening programme. Patients with a previous diagnosis of diabetic macular oedema, severe non-proliferative diabetic retinopathy, or proliferative diabetic retinopathy; previous laser treatment of the retina or retinal surgery; other non-diabetic retinopathy eye disease requiring referral to an ophthalmologist; or inability to have fundus photograph taken of both eyes for any reason were excluded. Deep-learning system-based interpretations of patient fundus images and referral recommendations were provided in real time. As a safety mechanism, regional retina specialists over-read each image. Performance of the deep-learning system (accuracy, sensitivity, specificity, positive predictive value [PPV], and negative predictive value [NPV]) were measured against an adjudicated reference standard, provided by fellowship-trained retina specialists. This study is registered with the Thai national clinical trials registry, TCRT20190902002.

Findings Between Dec 12, 2018, and March 29, 2020, 7940 patients were screened for inclusion. 7651 (96·3%) patients were eligible for study analysis, and 2412 (31·5%) patients were referred for diabetic retinopathy, diabetic macular oedema, ungradable images, or low visual acuity. For vision-threatening diabetic retinopathy, the deep-learning system had an accuracy of 94·7% (95% CI 93·0–96·2), sensitivity of 91·4% (87·1–95·0), and specificity of 95·4% (94·1–96·7). The retina specialist over-readers had an accuracy of 93·5 (91·7–95·0; $p=0\cdot17$), a sensitivity of 84·8% (79·4–90·0; $p=0\cdot024$), and specificity of 95·5% (94·1–96·7; $p=0\cdot98$). The PPV for the deep-learning system was 79·2 (95% CI 73·8–84·3) compared with 75·6 (69·8–81·1) for the over-readers. The NPV for the deep-learning system was 95·5 (92·8–97·9) compared with 92·4 (89·3–95·5) for the over-readers.

Interpretation A deep-learning system can deliver real-time diabetic retinopathy detection capability similar to retina specialists in community-based screening settings. Socioenvironmental factors and workflows must be taken into consideration when implementing a deep-learning system within a large-scale screening programme in LMICs.

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Introduction

Diabetes is a growing pandemic and major public health concern, with an estimated 463 million people affected worldwide in 2019.¹ A serious complication associated with diabetes is diabetic retinopathy, a leading cause of blindness in adults and a considerable burden on a patient's quality of life. The prevalence of diabetes is expected to increase to 700 million people by 2045,¹ with an expected concomitant increase in diabetic retinopathy incidence. To reduce the risk of blindness or visual impairment from diabetic

retinopathy, regular eye screenings are recommended for individuals with diabetes.²

Deep-learning systems have the potential to enhance and automate aspects of screening for diabetic retinopathy, and studies have examined their clinical applicability. Several retrospective and prospective studies sought to validate the use of deep-learning systems in diabetic retinopathy screening.^{3–11} These studies have shown robust diagnostic performance of deep-learning models across different ethnic populations. However, the

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Research in context

Evidence before this study

We searched Google Scholar, PubMed, Scopus, MEDLINE, and Web of Science for articles published in English from Oct 1, 2010, to Oct 1, 2020, using the keywords “diabetic retinopathy”, “automated diabetic retinopathy screening”, “diabetic macular edema”, “machine learning”, “artificial intelligence”, and “deep learning”. 20 studies have reported retrospective validation of deep-learning models in the detection of diabetic retinopathy, including our group. Most of the prospective clinical validations have been observational without affecting clinical care. No previously reported study applied a deep-learning system interventional as part of an existing large-scale screening programme in a low-income or middle-income country (LMIC).

Added value of this study

Our findings provide evidence for the feasibility of large-scale screening using a deep-learning system prospectively deployed within a national screening programme across multiple regions and sites in a middle-income country, Thailand. The deep-learning system enabled instantaneous diabetic retinopathy or diabetic macular oedema grading assessments and referral recommendations to patients with diabetes who typically would have had to wait weeks before receiving their screening results due to the scarce availability of eye doctors. In our previous retrospective study, the deep-learning system performed similarly to health-care personnel who typically screened and graded images in the Thai national

screening programme (ie, general ophthalmologists, nurses, and trained technicians). Building upon this, our study suggests that deep-learning system performance is maintained in a real-world setting and is similar to regional retina specialists. Qualitative insights and socioenvironmental factors to consider when introducing an artificial intelligence intervention in LMIC settings are also presented.

Implications of all the available evidence

An instantaneous screening tool with expert-level accuracy might be of use in resource-constrained settings, such as those with a shortage of eye specialists or high screening volumes. Furthermore, the deep-learning system might facilitate management of patients with diabetes by enabling earlier follow-up care and treatment through its real-time grading and detection of vision-threatening diabetic retinopathy. The findings presented here might inform efforts in other LMICs regarding potential technical and environmental challenges to address when implementing a deep-learning system into a large-scale screening programme or community care settings. Human grading and oversight can potentially be reserved for cases identified as referable or ungradable by a deep-learning system. This hybrid approach would enable the human grader to focus on more urgent cases, enhancing their ability to care for more patients. More research is required to understand the cost-utility of a deep-learning system applied in a national screening programme in a middle-income country.

adoption of artificial intelligence (AI) technology is dependent not on performance alone but also on integration and acceptability in real-world clinical workflows. Most prospective analyses of deep-learning systems in real-world settings have been observational, in which deep-learning systems did not affect clinical care or validate feasibility of real-world workflow integration.^{7–12} Deep-learning systems as an intervention at the point-of-care have been reported in some pilot studies with fewer than 200 patients, but the feasibility and performance of deep learning in large-scale deployments have not been reported, especially in low-income and middle-income countries (LMICs).

The burden of diabetes disproportionately affects LMICs, which have resource and manpower constraints.¹³ We focused on Thailand, a middle-income country with an estimated 4·5 million patients with diabetes.¹⁴ Thailand has a large-scale nationwide diabetic retinopathy screening programme that enables non-physician health-care personnel (mainly nurses) to manually screen eyes in primary care clinics and community hospitals. Although the programme successfully provides access to screening and treatment to a large Thai population, several challenges remain. First, programme resources are limited: this affects screening uptake, which

averages 50% of patients with diabetes, and falls short of the national target of 60%. Second, trained screening personnel have no higher than 85% accuracy in reading retinal images for determining eligibility for referrals to ophthalmologists;¹⁵ this reflects both missed (false negative) and unnecessary (false positive) referrals. Third, when a second opinion is sought, patients receive their diagnosis weeks to months after their eye screening visit, potentially effecting patient engagement and adherence. Lastly, there is poor understanding of how closely patient follow-up and referrals for treatment are tracked. These challenges are not unique to the Thai screening programme and are commonly observed in diabetic retinopathy screening programmes worldwide.

In these settings, a deep-learning system might be valuable in increasing screening efficiency and accuracy. Our deep-learning system has previously been tested and validated on independent datasets.^{3,5,10} In our retrospective study, compared with regional graders (general ophthalmologists, nurses, and technicians) in the Thai screening programme, the deep-learning system had significantly higher sensitivity (0·97 vs 0·74; $p<0\cdot001$) and a slightly lower specificity (0·96 vs 0·98; $p<0\cdot001$) for detecting referable diabetic retinopathy.³ We aimed to assess the deep-learning system's performance in detecting

vision-threatening diabetic retinopathy and diabetic macular oedema against a reference standard based on retina specialist adjudication. In addition, the deep-learning system performance was compared with regional retinal specialists who over-read all cases screened by the deep-learning system.

Methods

Study design and participants

This was a prospective cohort study done at nine primary care centres (eight community hospitals in rural settings and one primary care clinic in an urban tertiary care hospital), as part of the Thai national diabetic retinopathy screening programme. These sites were located across three regions: central Bangkok, Chiang Mai, and Pathum Thani, Thailand.

Patients with diabetes who were listed on the national diabetes registry, aged 18 years or older, able to have a fundus photograph taken for at least one eye, and due for screening as per the Thai Ministry of Public Health guidelines were eligible for inclusion. On designated screening days, eligible patients registered at the clinic were provided with information about the study. Patients who consented to participate were assigned a unique case identifier for pseudonymisation. Patients with a previous diagnosis of diabetic macular oedema, severe non-proliferative diabetic retinopathy, or proliferative diabetic retinopathy; previous laser treatment of the retina or retinal surgery; other non-diabetic retinopathy eye disease requiring referral to an ophthalmologist; or inability to have fundus photo taken of both eye for any reason were excluded.

This study was approved by the Ethics Review Committee at all participating hospitals. Patients gave written informed consent allowing their retinal images to be used in the study. The study is registered with the Thai Clinical Trials Registry. Patient enrolment started after Ethics Review Committee approval; trial registration occurred after 138 patients were recruited due to technical issues in document registration and delays in submission and review by the Thai clinical trials registration organisation. CONSORT-AI reporting guidelines¹⁶ were followed in this study.

Development and clinical validation of our deep-learning systems has been previously described using datasets across different patient populations and fundus cameras, including a retrospective study in the Thai national screening programme.^{3,5,17} This prospective study applied the latest Conformité Européenne-marked version (version 2) of the algorithm (indicates conformance with European Union product legislation).¹⁸ The deep-learning system provides an output of a 5-point diabetic retinopathy severity grade and presence or absence of diabetic macular oedema, according to the International Clinical Diabetic Retinopathy severity scale,¹⁹ and gradability of both diabetic retinopathy and diabetic macular oedema. The deep-learning system was

trained to classify images as ungradable using examples of images interpreted as ungradable by human graders, rather than based on manually specified features, such as blurriness. The deep-learning system's ungradable classifications are associated with several factors including but not limited to poor image quality (exposure or artefacts), clarity (eg, blurry images), media opacities, or incorrect field of view.

Procedures

All participating patients were screened prospectively by the deep-learning system. The study aimed to integrate deep-learning systems into existing workflows with minimal disruption. Camera operators used existing fundus cameras at each site: Topcon Maestro 3D OCT-1, Topcon TRC-NW300, Nidek AFC-230, Nidek AFC-210, and Nidek AFC-300 (appendix 2 p 12). Non-mydratric, 45°, macular-centred, single-field, colour fundus photographs were taken of both eyes (or one eye if not possible for both). Pupils were dilated only if required, except at Chomthong Hospital, Chiang Mai, Thailand, in which standard practice was to dilate all pupils.

Images taken were uploaded to the cloud via mobile WiFi hotspots for processing by the deep-learning systems using case number as the sole identifier. The deep-learning system provided a real-time recommendation for referral for comprehensive ophthalmology follow-up on the basis of severe non-proliferative diabetic retinopathy or above in either eye; diabetic macular oedema in either eye; or ungradable image in either eye. Patients were also referred for visual acuity based on measurement with a pinhole: visual acuity of 20/70 or worse, or worsening vision defined as reduced visual acuity by two or more lines from previous visits. The deep-learning system was not intended to alter the sites' existing referral criteria, so these referral criteria reflect an update (before trial commencement) relative to an earlier protocol that stated the visual acuity threshold as 20/40 and the diabetic retinopathy severity threshold as moderate non-proliferative diabetic retinopathy. The nurses conveyed the referral recommendation to patients based on the deep-learning system and visual acuity results entered into a custom user interface. Patients who were not referred were scheduled for the next screening within 6–12 months. There were no repeat screenings in the study (unique visits captured only).

After screening completion, participating sites logged data into a web-based application created for the study to collect and manage study-related deidentified patient data. The baseline patient characteristics collected included patient demographics and risk factors. The web-based application also enabled remote over-reading and referral tracking. Because there was no systematic tracking of referrals within the Thai diabetic retinopathy screening programme or electronic medical record, we implemented a customised tracking method for the study. Referral

For more on the trial see <https://www.thaiclinicaltrials.org/show/TCTR20190902002>

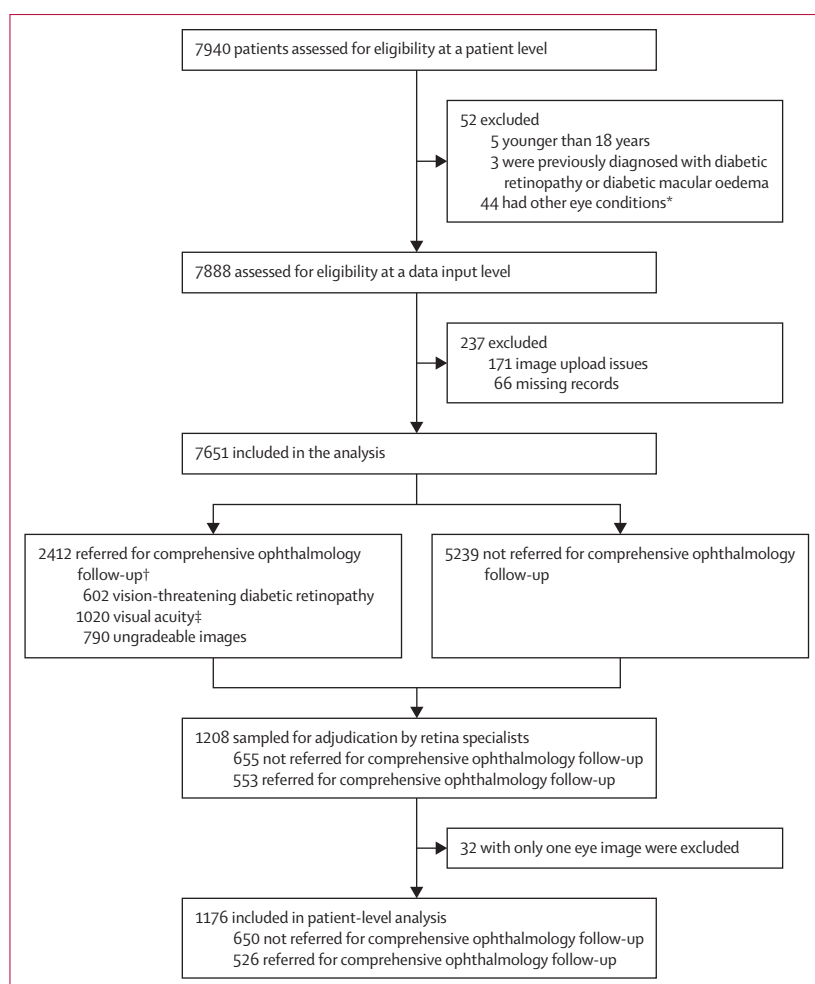


Figure 1: Study profile

*Examples of other eye conditions included: glaucoma, age related macular degeneration, epiretinal membranes, branch retinal artery exclusion, and central retinal vein occlusion. †Referrals for vision-threatening diabetic retinopathy and ungradable images were based on the deep-learning system interpretation whereas visual acuity was measured by the nurses. Reasons for referral are presented here based on clinical priority: vision threatening diabetic retinopathy, low visual acuity, and ungradable image. In other words, if a patient had two reasons to be referred: one eye with vision threatening diabetic retinopathy and the other eye with an ungradeable image, the referral reason was counted as vision threatening diabetic retinopathy. ‡Of the patients referred for visual acuity, only six patients were referable for decreased visual acuity of two lines or more in at least one eye. Of those six, one patient had both eyes with 20/70 visual acuity, and a two line decline since the previous screening session. The remaining patients were referred for visual acuity of 20/70 or worse in one eye.

See Online for appendix 2

tracking was dependent on the patient presenting the study-marked referral slip issued by the screening clinic at the referral hospital. This allowed the referral hospital to recognise study participants and subsequently notify the study co-ordinator to record referral completion within the web-based application. Because this process was partly paper based, incomplete recording of referral completion was possible; therefore, an underestimation of actual referral rates within the study could have occurred.

At study completion, patient records were reviewed to ensure ineligible patients were not entered into the study. Patients with any of the exclusion criteria (as indicated by the over-readers) were excluded from the analysis.

Every image analysed by the deep-learning systems was also over-read for safety by regional retinal specialists (KH [10 years' post-certification experience] and AK [3 years' post-certification experience]). Over-reading checked for referable cases potentially missed by the deep-learning systems (false negatives). Regional over-readers were masked to the deep-learning system results and provided an independent assessment for diabetic retinopathy, diabetic macular oedema, and additional suspected pathologies.

If the regional over-readers determined a case required a referral after the deep-learning system deemed the case not referable, an automated notification was sent to the screening site and study co-ordinator. The local nurses could then contact the patient to arrange a referral.

A reference standard was created to compare the performance of the deep-learning system with the regional over-readers. This was done by ground truth adjudication of individual eye images for diabetic retinopathy severity and referable diabetic macular oedema. Because the large sample size rendered exhaustive adjudication of all images infeasible, we applied stratified sampling and inverse-probability weighting for adjudication, using a strategy described previously (appendix 2 p 4).³

All images sampled for adjudication were reviewed by panels of three US board-certified retina specialists, drawn from a pool of 11 specialists (mean of 5 years of post-certification experience). Each image was initially assessed independently by each panel member. In case of disagreement, each of the previous assessments and comments from each adjudicator were reviewed by one adjudicator at a time in a round-robin fashion until all panel members agreed upon gradability, and, if the image was gradable for either condition, diabetic retinopathy severity and referable diabetic macular oedema assessments. Adjudicators were masked to the identity of other adjudicators and to deep-learning system interpretations. There was no overlap between panel adjudicators and regional over-readers in the study.

Deep-learning system performance was assessed at the individual eye and patient levels. The patient-level reference standard was derived by taking the maximum severity across eyes (eg, a patient with mild non-proliferative diabetic retinopathy in one eye and proliferative diabetic retinopathy in the other eye would be considered to have proliferative diabetic retinopathy). The clinical protocol is available in appendix 2 (pp 13–14).

Outcomes

Primary endpoints for evaluation of deep-learning system performance were specified before initiation of the study. These included accuracy, sensitivity, specificity, and positive predictive value (PPV) and negative predictive value (NPV) of diabetic retinopathy and diabetic macular oedema diagnoses. Metrics were assessed by comparing deep-learning system predictions to an adjudicated reference standard, described below.

The primary clinical endpoint was vision-threatening diabetic retinopathy, defined as either severe non-proliferative diabetic retinopathy, proliferative diabetic retinopathy, or referable diabetic macular oedema. This was used as the referral criterion for patients in this study and for sample size calculations.

As secondary measures, we assessed performance of the deep-learning system for detecting moderate or worse diabetic retinopathy or referable diabetic macular oedema, which are both used as a referral criteria in other screening programmes,^{9,11} and image gradability for diabetic retinopathy and diabetic macular oedema. We present exploratory analyses of additional endpoints, such as all cause referable performance, detection of different diabetic retinopathy severity thresholds, and performance stratified by camera model (which is correlated with site). To contextualise the performance of the deep-learning system, we compared deep-learning system performance metrics (accuracy, sensitivity, specificity, positive predictive value, and negative predictive value) with the regional over-readers who also reviewed each image. Additionally, we did prespecified qualitative surveys to assess staff satisfaction with deep-learning system-based screening workflows at study completion.

Statistical analysis

Sample size was estimated by a power analysis done before study recruitment, using a binomial approach.^{3,20} The power analysis used the following parameters: 0·05 as the estimated proportion of the screened population at risk for vision-threatening diabetic retinopathy (based on observed diabetic retinopathy progression rates²¹) and a tolerated margin of error of 0·10. This produced an estimate of 7299 patients, which was used along with a dropout estimate of 4–5% to set an approximate recruitment target of 7600 (7603 for 4% dropout) for the study.

To account for the sampling process used for adjudication, we applied inverse probability weighting, weighting assessments from each patient in inverse proportion to its likelihood of being sampled for adjudication.²² The resulting metrics thus estimate performance of the deep-learning system and over-readers on the entire screening population. We calculated 95% CIs using a non-parametric bootstrap approach with 1000 bootstrap samples. All prespecified and post-hoc analysis p values were calculated using two-sided permutation tests.

Role of the funding source

Google codeveloped the study design with PR and analysed and interpreted the data.

Results

From Dec 12, 2018, to March 29, 2020, 7940 patients were assessed for eligibility of whom 7651 (96·3%) patients

	All patients (n=7651)	Adjudicated sample* (n=1176)
Mean age (years)	59·9 (10·3)	59·8 (10·4)
Sex		
Female	4830 (63·1%)	746 (63·4%)
Male	2818 (36·8%)	430 (36·6%)
Other	3 (<0·1%)	0
Diabetes		
Type II	7219 (94·4%)	1140 (96·9%)
Type I	422 (5·5%)	35 (3·0%)
Gestational	2 (<0·1%)	0
Prediabetes	7 (0·1%)	0
Unknown	1 (<0·1%)	1 (<0·1%)
Mean diabetes duration (years)	7·6 (6·2)	7·6 (6·2)
Mean systolic blood pressure (mmHg)	130·4 (13·9)	132·1 (14·6)
Mean diastolic blood pressure (mmHg)	74·4 (9·9)	74·1 (10·1)
Mean HbA1c concentration	7·5 (1·9)	8·0 (2·1)
Smoking status		
Yes	329 (4·3%)	54 (5·6%)
No	7231 (94·5%)	1104 (93·9%)
Not available	91 (1·2%)	18 (1·5%)
Diabetic retinopathy severity (from the deep-learning system)		
Ungradable	1131 (14·8%)	201 (17·1%)
No diabetic retinopathy	4928 (64·4%)	263 (22·4%)
Mild non-proliferative diabetic retinopathy	660 (8·6%)	85 (7·2%)
Moderate non-proliferative diabetic retinopathy	747 (9·7%)	494 (42·0%)
Severe non-proliferative diabetic retinopathy	86 (1·1%)	66 (5·6%)
Proliferative diabetic retinopathy	99 (1·3%)	67 (5·7%)
Diabetic macular edema (from the deep-learning system)		
Ungradable	1105 (14·4%)	171 (14·5%)
No diabetic macular oedema	5969 (78·0%)	629 (53·5%)
Referable diabetic macular oedema	577 (7·5%)	376 (32·0%)

Data are n (%) or mean (SD). *Enriched for disagreements between the deep-learning system and over-reader, resulting in a higher proportion with findings.

Table 1: Baseline characteristics of eligible participants and adjudicated sample set

were included in the analysis (figure 1). Due to clinical constraints, the exact numbers of patients not meeting this inclusion criteria were not available. Figure 1 presents available information on the screened patients. 4830 (63·1%) women and 2818 (36·8%) men—mean age 59·9 years (SD 10·3)—were recruited (table 1). The median time to over-read after initial deep-learning system read was 10 days (95% CI 2–49); 2357 (35%) of 7578 patients (for whom over-read time was reported) received an over-read in a week. 7231 (94%) participants self-reported as non-smokers, which was higher than the reported prevalence of this group in Thailand (range 80–86%).^{23,24} The characteristics of patients in the

adjudication set was similar to that of study-eligible patients, with the exception of a deliberately increased proportion of patients with referable disease to improve the statistical efficiency of adjudication (table 1; appendix 2 p 4). 2412 (31.5%) patients were referred for comprehensive ophthalmology follow-up (figure 1). We were able to track referral adherence for a subset of patients. Of the 222 patients who presented for referral, the most common reason was vision-threatening diabetic retinopathy (141 [64%] patients), followed by visual acuity (60 [27%] patients) and ungradable images (21 [9%] patients).

2384 (15.6%) of 15 270 images from 1208 (15.6%) of 7651 patients were adjudicated. Of these patients, 32 (0.4%) patients had only one eye adjudicated (either because the patient had one eye or the other eye could not be imaged). Patient-level analyses were confined to 1176 (97.4%) of 1208 patients for whom both eyes were adjudicated.

Our deep-learning system had high performance on all the prespecified primary endpoints (table 2). Across

patients, the deep-learning system had an accuracy of 94.7 (95% CI 93.0–96.2), sensitivity of 91.4% (87.1–95.0) and specificity of 95.4% (94.1–96.7). Compared with the deep-learning system, the retina specialist over-readers had an accuracy of 93.5% (91.7–95.0; $p=0.17$), a sensitivity of 84.8% (79.4–90.0; $p=0.024$), and specificity of 95.5% (94.1–96.7; $p=0.98$).

We also examined the PPVs and NPVs of the deep-learning system (table 3; appendix 2 p 11). Of patients whose images were interpreted as non-referable by the deep-learning system, the over-readers agreed with 5613 (96%) of the 5865 classifications made by the deep-learning system; the adjudication panel agreed with 579 (93.7%) of 618 classifications. This suggests a high NPV for screening recommendations. Of the 5865 (76.7%) of 7651 patients not referred by the deep-learning system, only 104 (1.8%) patients were later determined by the over-readers to require a referral for vision-threatening diabetic retinopathy. By contrast, of the 602 (7.8%) of 7651 patients referred for vision-threatening diabetic retinopathy by the deep-learning system, 387 (64.3%) were interpreted as referable by over-readers, and 236 (57.7%) of 409 patients were similarly classified by the adjudication panel. This low observed rate might have been influenced by the relatively low prevalence of vision-threatening diabetic retinopathy in our cohort. When computing reweighted PPV values to reflect the screening population, PPV for vision-threatening diabetic retinopathy was 79.2% (appendix 2 p 11). Qualitative inspection of false positive and false negative examples suggested that many of these examples involved ambiguity between pathology diagnostic of diabetic macular oedema (eg, hard exudates) and not being diagnostic of the disease (eg, drusen). Many of the discordances in either direction tended to involve over-calling or under-calling of diabetic macular oedema specifically (appendix 2 pp 2, 8).

Because referrals might incur costs, such as travel or time off work for patients,²⁵ comparison of patient referrals due to ungradable images versus referable diabetic retinopathy or diabetic macular oedema is pertinent. To better understand this question, we analysed how often the deep-learning system and over-readers agreed with the adjudicated labels for image

	Patients	Deep-learning system	Retina specialist over-readers	Difference between DLS and over-readers (95% CI)	Non-equivalence p value (permutation test)
Patient-level metrics					
Accuracy	926 (100%)	94.7 (93.0 to 96.2)	93.5 (91.7 to 95.0)	1.2 (-0.4 to 3.0)	0.17
Sensitivity	255 (28%)	91.4 (87.1 to 95.0)	84.8 (79.4 to 90.0)	6.6 (2.0 to 11.2)	0.02
Specificity	671 (72%)	95.4 (94.1 to 96.7)	95.5 (94.1 to 96.7)	0 (-1.6 to 1.6)	0.98
Eye-level metrics					
Accuracy	987 (100%)	95.8 (94.5 to 96.9)	94.8 (93.4 to 96.2)	1.0 (-0.4 to 2.4)	0.20
Sensitivity	164 (17%)	91.3 (85.1 to 96.4)	89.6 (83.3 to 94.7)	1.7 (-3.7 to 6.8)	0.43
Specificity	823 (83%)	96.3 (95.1 to 97.4)	95.4 (94.1 to 96.7)	0.9 (-0.5 to 2.4)	0.24

Data are n (%) or estimate (95% CI). p values were calculated with permutation test. Metrics were adjusted via inverse probability weighting to account for stratified sampling (appendix 2 p 4). Metrics were computed on all cases considered gradable by both the deep-learning system and over-reader; all-cause referable analysis is reported in appendix 2 (p 10).

Table 2: Performance of the deep-learning system and regional over-readers against ground truth adjudication for vision-threatening diabetic retinopathy

	Over-readers			Adjudication panel		
	No referral	Referral	Ungradable	No referral	Referral	Ungradable
No referral	5613/5865 (96%)*	104/5865 (2%)	148/5865 (3%)*	579/618 (94%)*	28/618 (5%)	11/618 (2%)
Referral	189/602 (31%)	387/602 (64%)*	26/602 (4%)	134/409 (33%)	236/409 (58%)*	39/409 (10%)
Ungradable	739/1184 (62%)	34/1184 (3%)	411/1184 (35%)	62/181 (34%)	5/181 (3%)	114/181 (63%)*

Data are n/N (%). Data are for the adjudication set without sampling correction. To focus on diabetic retinopathy interpretations, these categorisations do not consider other possible referral reasons such as visual acuity. Adjudication counts reflect the subset of patients whose images were adjudicated; the raw proportions reported do not account for adjudication sampling, and do not directly reflect positive predictive values or negative predictive values (appendix 2 p 11). *Most common over-reader or adjudication panel interpretation for each category of DLS interpretation.

Table 3: Agreement between the deep-learning system and regional over-readers and adjudication panel

gradability. Of the images marked as ungradable by the deep-learning system, over-readers often disagreed, marking only 411 (34.7%) of 1184 such images as ungradable. By contrast, 739 (62.4%) of 1184 of these images were estimated to be ungradable by the adjudicated panel (table 3). Compared with the adjudicated reference standard, the deep-learning system showed high performance on referrals for ungradability, with high overall accuracy (90.4% [95% CI 87.8–92.7]; $p=0.028$), sensitivity (88.2% [81.1–94.2]), and specificity (90.8% [88.0–93.2]; appendix 2 p 10). By contrast, regional over-readers had similar overall accuracy (86.3% [88.0–93.2]) and specificity (94.6% [92.5–96.5]; $p=0.028$), but their sensitivity was significantly lower (34.7% [24.8–44.9]; $p=0.0040$) compared with the deep-learning system, due to calling fewer images as ungradable.

We did qualitative and quantitative post-hoc analyses to better understand ungradable differences between the deep-learning system and regional over-readers (figure 2; appendix 2 pp 7–8). 31 (29.3%) of 104 images marked ungradable by adjudication were also marked ungradable by both the deep-learning system and over-readers; these often included multiple areas of low visibility, low image quality, or blurriness (figure 2A). Over-readers marked such images as gradable more often than the deep-learning system (figure 2B–D; table 3). Visual inspection of examples suggest that of the ungradable cases marked by over-readers as gradable, a range of disease severity was visible, with similar numbers of cases having and not having referable disease (appendix 2 pp 7–9). This suggests that disagreements around gradability might not stem from a single point alone.

Deep-learning system performance was similar across different camera models, which was associated with the respective sites in which each camera model was used. However, image gradability—as determined by the adjudication panel—varied (appendix 2 p 12). The highest rates of ungradable images (224 [39.1%] of 573 by deep-learning system vs 43 [7.5%] of 573 by adjudication panel; after reweighting to account for sampling) were observed with the Nidek AFC-230 model used at Lamlukka Hospital, Pathum Thani, Thailand (appendix 2 p 12). Technical issues with the camera resulted in poor quality images and were a contributing factor to the higher ungradability at Lamlukka. The lowest ungradable rate (8.41%) was observed with the TopCon TRC NW330, used at three sites: Doi Saket, Chomthong, and Phrao, Chiang Mai, Thailand. The deep-learning system and regional over-readers showed similar trends in ungradability rates across camera models; however, the over-readers tended to mark substantially fewer images as ungradable (1.17–13.17%) compared with the deep-learning system (10.78–39.09%) or adjudication panel (3.69–30.03%).

The 12 staff members who screened participants completed qualitative surveys to assess satisfaction and

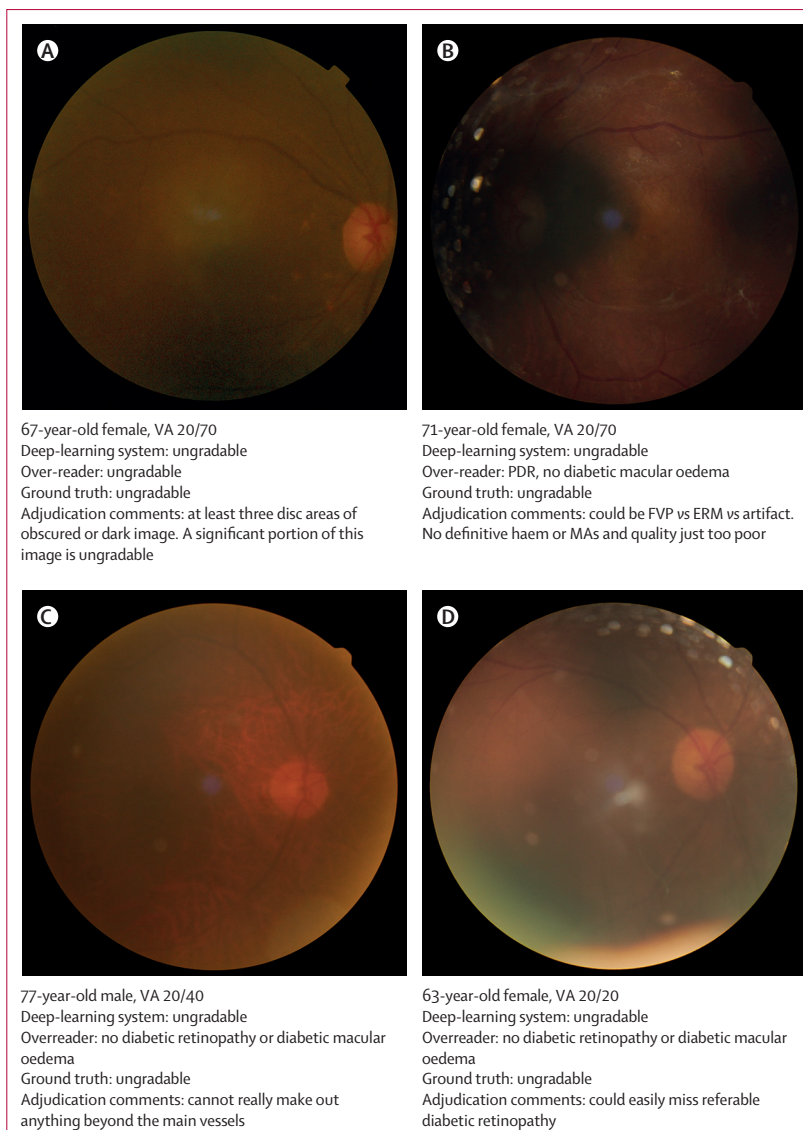


Figure 2: Examples of eye images to examine ungradability differences between deep-learning system and over-reader

ERM=epiretinal membrane. FVP=fibrovascular proliferation. Haem=haemorrhage. MAs=microaneurysms. PDR=proliferative diabetic retinopathy. VA=visual acuity.

trust in the deep-learning system-based screening being introduced for the first time. The nurses adapted well to the deep-learning system-based workflow and cited an overall positive experience and confidence in the system (appendix 2 p 3). From interviews and survey responses, the nurses indicated that they liked the benefit of instantaneous results from the deep-learning system and its ability to aid in referral decision-making and, in some cases, task-shifting to less trained staff. Challenges cited were related to images deemed ungradable by the deep-learning system and additional time and steps required with uploading images to a separate system for analysis.

Discussion

This study integrated a deep-learning system into routine screening workflows at the point-of-care in primary care centres as part of a national screening programme. The specific goals were to evaluate the feasibility of deploying a deep-learning system and testing its performance prospectively in clinical settings. The study also aimed to qualitatively assess socioenvironmental factors related to real world implementation of deep-learning system in LMICs.

To our knowledge, this prospective study is the largest real-world validation to-date of a deep-learning system for eye screenings, having screened more than 7000 patients across both urban and rural care settings. The deep-learning system had similar accuracy to Thai retina specialist over-readers with significantly higher sensitivity and similar specificity for vision-threatening diabetic retinopathy and secondary endpoints, such as moderate or worse diabetic retinopathy and diabetic macular oedema (appendix 2 pp 10–11). We have reported robust performance of our deep-learning system versus human graders in previous studies across different patient populations.^{3,5,10} Combined, these studies build evidence for the generalisability of the deep-learning system and its feasibility in scaling screening programmes while maintaining standard of care.

In addition to accurate reads, our deep-learning system also provided screening results in real-time. This was a substantial improvement for patients who previously had to wait weeks to months before receiving their screening results and referral recommendations. Nurses felt empowered to provide better guidance to their patients with deep-learning system-enabled real time results and referral recommendations.²⁵ Nurses also cited the potential for task-shifting, following which less trained personnel could screen patients and allow the more experienced nurses to focus on other aspects of patient care.

Emerging evidence from other studies suggests that real-time screening results provided by the deep-learning system at the point-of-care correlate to improved referral adherence.^{26,27} However, this analysis was outside the scope of our investigation. We observed previously unknown insights about patient referral adherence. Our referral tracking showed that patients who attended their referrals were most commonly referred for referable diabetic retinopathy or diabetic macular oedema, compared with low visual acuity, and ungradable images. This might be due to the perceived higher urgency for treating vision loss from patients referred due to diabetic retinopathy and diabetic macular oedema and the lower urgency for visual acuity that declines more gradually.

We also assessed the effect of a deep-learning system on screening workflows. The challenges with integrating new AI technology into existing systems need to be carefully considered to create a seamless workflow. Currently in the screening programme, patient data is

managed entirely with paper-based records, with no system to track completed patient referrals. The absence of a digitised system slowed down the deep-learning system-based screening process (eg, manual image upload was necessary) and hindered measuring its effect on referral adherence rates (ie, no historical data to compare with). For future scalable implementations, health-care infrastructure with digitised screening systems and interoperability across institutions and departments is an important consideration.

A second challenge was the number of referrals by the deep-learning system due to ungradable images. Ungradable deep-learning system interpretations were mostly associated with poor image quality resulting from operator-dependent factors, camera-related issues (as seen in Lamlukka which had high ungradability), or ocular media opacities (usually cataracts). 64% of images considered ungradable by the deep-learning system were also considered ungradable by the adjudication panel, higher than the 35% agreement rate with over-readers. This difference likely arose because the panel had more time to review and deliberate over the gradings until consensus was achieved, whereas the over-readers individually graded each image under real-world clinical time constraints (within 7 days), in a manner consistent with their routine practice. Furthermore, although over-readers followed standard guidelines (International Clinical Diabetic Retinopathy severity scale), specific criteria for evaluation of ungradability were not provided. Therefore, over-readers might have had a different comfort level for uncertainty in evaluating diabetic retinopathy and diabetic macular oedema status clinically. Whether providing a defined set of criteria for ungradable image evaluation would have affected the agreement between over-readers and panel adjudicators is unknown. A helpful future feature for the deep-learning system might be to provide the reason for ungradability (eg, poor image quality, camera-related artefacts, and presence of media opacities) to alleviate much of the uncertainty related to the cause of an ungradable image and reduce unnecessary referrals. Alternatively, the threshold for calling a case ungradable could potentially be customised to fit local resource constraints and practice patterns. Human-centred approaches, including camera operator training and camera maintenance,^{28,29} should also help reduce ungradability rates.

The third key challenge was that slow internet connectivity occasionally delayed image upload and receipt of results. As part of study on-boarding and setup, internet connectivity issues were mitigated by providing each site with a pocket WiFi device which helped ensure a consistent internet connection. Nevertheless, better image compression or an offline deep-learning system might help in this regard, particularly for community hospitals with large patient volumes and mobile screening settings.

Real-world use of a deep-learning system in LMICs needs to take into account the available resources to ensure sustainability and cost-effectiveness of the endeavour. A health economic analysis within the Singapore national screening programme determined that a semi-automated approach that maintained human read for cases identified as referable by deep-learning system resulted in a 20% reduction in cost versus a fully automated model or one reliant solely on human graders.³⁰ On the basis of our results, a semi-automated approach can reduce unnecessary referrals by diverting only positive or ungradable cases (per the deep-learning system) to be over-read by eye specialists, more efficiently using experts' time. Future studies on the cost-effectiveness of such a semi-automated approach within the Thai national screening programme will provide additional insights.

This study had some limitations. The absence of electronic medical records or systematic referral tracking within the screening programme required us to implement our own custom tracking process within the study. Although the study-reported referral adherence rates might be underestimated, they provide insights on screening programme effectiveness and the type of patients more likely to complete their referral visits. Furthermore, the deep-learning system used in our study was trained to grade diabetic retinopathy severity on a 5-point scale. This enables broad applicability in both LMICs and high-income countries because referral criteria or trade-off between sensitivity and specificity could be customised on the basis of standard practises and cost implications for a screening setting. Additionally, visual acuity is an important measurement evaluated as part of referrals but one that the model does not account for. In other similar diabetic retinopathy screening programmes, it would be important to consider how visual acuity thresholds can be incorporated into referral generation logic (as we did in this study using a custom user interface). Although most diabetic retinopathy screening pathways refer out-patients with non-diabetic retinal conditions to other referral pathways,^{15,31–33} exploration of a deep-learning system that detects other conditions would be of interest. The development of deep-learning models that are capable of detecting multiple retinal conditions is an important area of active research.^{34,35}

In conclusion, this study provides more evidence that a deep-learning system can be deployed for large-scale diabetic retinopathy screenings with specialist-level accuracy in community settings in which screenings are typically managed by non-physician health-care personnel. To increase patient access to care, screening programmes might explore use of a deep-learning system, carefully evaluated for their settings, to extend screening to more centres.

Contributors

PR, BL, SV, KW, FH, LP, and DRW conceived the study and designed the protocol. VN and RT managed the study and data collection. RT, YL, and RS analysed the data analysis and drafted the manuscript. MS aided

data analysis. KH and AK over-read the retinal images. RL managed adjudicated data collection. JC provided access to the research tool used for patient data management in the study. RR critically reviewed the manuscript. JC and RR verified the data. All authors had access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

RT, FH, LP, DRW, RS, KW, SV, and YL are Google employees and own Alphabet stock. All other authors declare no competing interests.

Data sharing

Data might be made available for researchers with ethical approval from the diabetic retinopathy programmes of Rajavithi Hospital, Bangkok, Thailand; Khlong Luang, Nongseu, Lamukha, Thanyaburi Hospitals, Pathum Thani, Thailand; and San Patong, Phrao, Chomthong, and Doi Saket Hospitals, Chiang Mai, Thailand. Interested parties should contact PR (paisan.trs@gmail.com). Those interested in using or evaluating the deep-learning system should contact SV (svirmani@google.com).

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